

510 Python Apartment Finder Project

Christian Chinchilla

1. Motivation / Rationale

There are many factors that must be taken into consideration when hunting for apartments. Apartment websites usually focus only on price and amenities, while other important information such as commute time or bike accessibility must be searched separately. I plan to scrape data and integrate it for easy analysis. The goal of this project was to create a data pipeline that integrates apartment listing data with external API data to provide a more complete comparison of neighborhoods. The project answers the following questions:

- Which neighborhoods have the highest and lowest average rent?
- Which cities provide the best value in terms of price per square foot?
- How do commute times compare between neighborhoods?
- Which neighborhoods are the most walkable or bike friendly?

This project allowed me to combine several different Python techniques into one workflow, including scraping apartment data, cleaning it, integrating APIs, and visualizing the results.

2. Description of Data Sources

The project combines three separate data sources:

Craigslist Apartment Listings

Apartment listing data was scraped from Craigslist apartment search pages using Python requests and BeautifulSoup.

Example source URL: <https://losangeles.craigslist.org/search/apa>

The scraper collects: apartment title, city, monthly rent, square footage, number of bedrooms, and listing URL. The project scrapes more than 30 listings from each city, those being Downtown Long Beach, Cerritos, Redondo Beach, and Malibu to be able to give insights on different neighborhoods.

The scraper used command-line arguments to dynamically search different bedroom counts. This is provided via command line arguments to the script and will search for that number of bedrooms

Example: `python src/get_data.py 2 # 2 being the number of bedrooms.`

The scraper also removed duplicate apartment listings based on title and price combinations.

OpenRouteService API

API source: <https://api.openrouteservice.org/>

The project used the OpenRouteService API to calculate commute times from each city to USC and my own work location. The API returned estimated driving commute times in minutes. The final commute dataset included both USC commute times and work commute times.

WalkScore API

API source: <https://www.walkscore.com/professional/api.php>

The API scores were used to evaluate neighborhood accessibility and transportation convenience. The WalkScore API was also used to retrieve a walk score and bike score.

Challenges Encountered

Several challenges occurred during development were that Craigslist pages sometimes returned inconsistent HTML structures. Some apartment listings needed additional attention to try and extract accurate square footage and others did not include it. Duplicate listings appeared across searches and had to be removed.

3. Integrated Data Model

The project integrates apartment listing data with commute and walkability datasets. The main table is the Apartments table, which stores listing information scraped from Craigslist.

Apartments – apartment_id, city, listing_title, price, beds, square_footage, listing_url

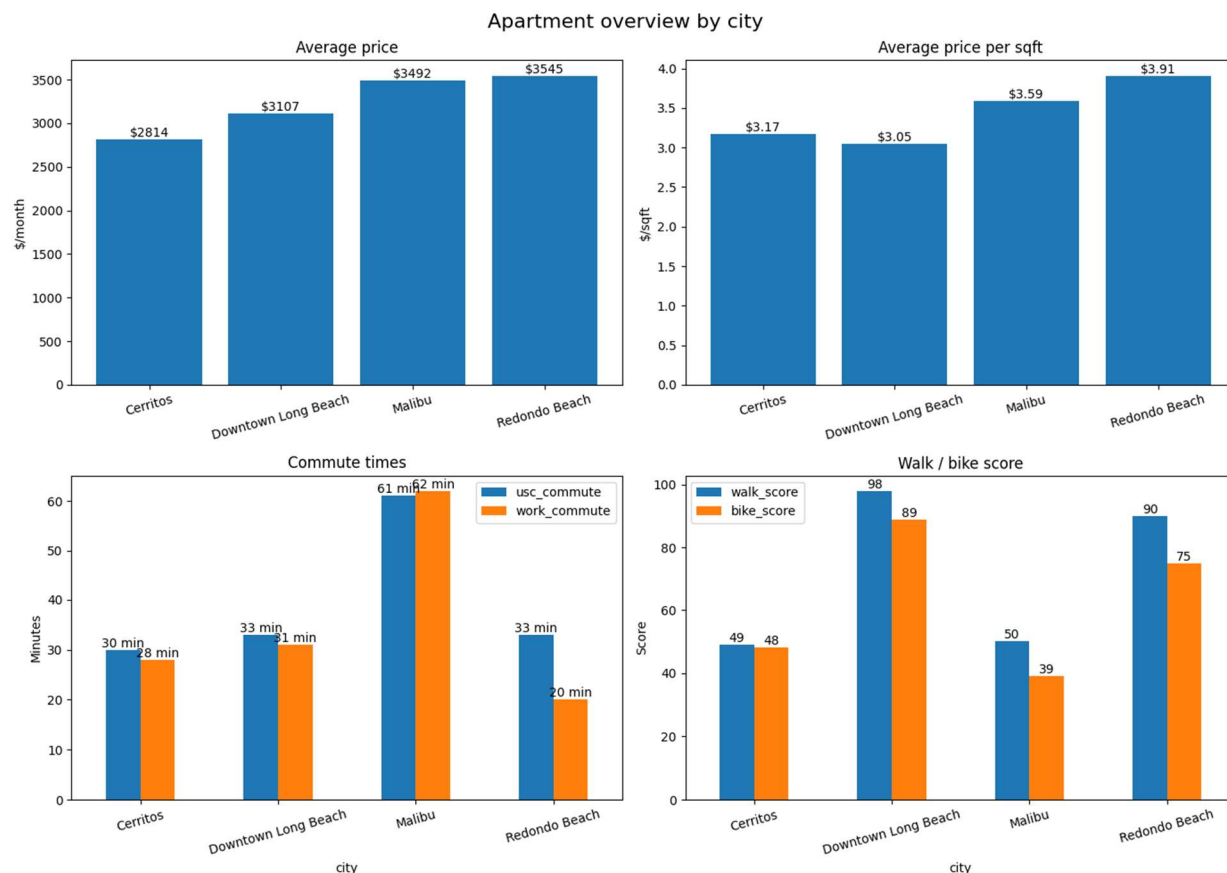
Walkscore – city, walk_score, bike_score

Commute – city, usc_commute, work_commute

This data is joined on the city column.

4. Analyses and Visualizations

The project used pandas and matplotlib for analysis and visualization.



Visualization 1 — Average Apartment Price

Findings: Malibu had the highest average rent (~\$3492/month). Cerritos had the lowest average rent (~\$2814/month). Redondo Beach also showed relatively high rental costs.

Visualization 2 — Average Price per Square Foot

Findings: Redondo Beach had the highest average price per square foot (~\$3.91/sqft). Downtown Long Beach had the lowest (~\$3.05/sqft). This shows that high prices often do not correlate with high value for square footage.

Visualization 3 — Commute Times

Findings: Malibu had the longest commute time (~60+ minutes). Cerritos had the shortest commute times (~30 minutes). Redondo Beach had the shortest work commute (~20 minutes). Given that I will only be commuting to USC for school it is more reasonable to choose an apartment with a lower commute time to my work location in El Segundo. These results demonstrate how commute convenience varies significantly across neighborhoods.

Visualization 4 — Walk and Bike Scores

Findings: Downtown Long Beach had the highest walk score (98). Redondo Beach also scored highly for both walking and biking. Malibu scored very low at 50.

This suggests that coastal areas with lots of typical urban features and high density of buildings are often more walkable.

5. Conclusions

The project demonstrated how combining multiple datasets creates a more useful and efficient apartment comparison system than relying on listing websites alone. Going through listings can also be tedious and lead to fatigue. A process like this one can help identify favorable apartments based off of key factors and then provides links in one source (apartments.csv) to be able to be opened after for further analysis. Malibu offered larger and more expensive apartments but had poor commute times and lower walkability. Downtown Long Beach provided extremely high walkability and strong transportation accessibility. Cerritos balanced affordability with relatively short commute times. Redondo Beach had high prices but also strong bike and walk scores. Each person has to balance out what is important to them and this project helps a user have that at a glance. Future integrations would probably include being able to add more cities, being able to filter based off wanted amenities, and cutting off listings that don't meet a price range. Overall, the project gave me experience working with real-world data collection, APIs, cleaning inconsistent data, and creating visualizations from the final dataset.